Range-Adaptive FMCW Radar: Real-Time Adaptation of RF and Baseband for Robust and Energy-Efficient Sensing

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***Abstract*— Static radar parameters fix the obtainable sensing range and force high RF power/gain, often wasting energy and requiring multiple modules to cover different ranges. We develop a closed-loop, range-aware adaptation framework that tunes RF front-end and ADC parameters on a per-frame basis from the detected range, thereby matching the detectable distance to the scene while dynamically scaling RF front-end performance. A lightweight adaptation pipeline refines detections, applies temporal smoothing, and maps the next-frame configuration using a pre-trained ML model under an energy-cost constraint. Implemented in closed loop on a commercial radar module, the approach reduces total energy by 33 to 69% without degrading detection performance. The end-to-end latency including signal processing and configuration upload, achieved a median of 21.76 ms, validating real-time operation.**

***Keywords*— adaptive algorithms, cognitive radar, closed loop systems, radar signal processing.**

I. Introduction

Conventional radar systems employ static parameter settings and are typically categorized as short-, mid-, or long-range, often requiring multiple radar modules to cover different detection ranges. For example, in automotive applications, long-range radars are used for adaptive cruise control while short-range radars for parking assistance [1], [2]. Similarly, in surveillance, short-range modules are used for obstacle avoidance, whereas long-range modules are employed for terrain mapping [3], [4]. However, this multi-module architecture increases system cost, hardware redundancy, and processing complexity.

Cognitive radar can address this limitation by dynamically adjusting system parameters through adaptive feedback across diverse operating conditions [5]. Previous works have explored adaptive waveform control [6], [7] or RF front-end tuning via look-up table (LUT) [8], [9], but these approaches often suffer from high latency and adaptation flexibility. More recently, artificial-intelligence based methods [10], [11] have been investigated to adapt radar parameters in real-time scenarios, yet they primarily focus on algorithmic optimization or rely on simulation using public datasets without real closed-loop hardware implementation.

To demonstrate the real-time feasibility of cognitive radar, we propose an adaptive framework that dynamically adjusts the hardware parameters of radar module according to the detected target range frame-by-frame. Based on detection results from the previous frame, the system selects optimized parameters for the upcoming frame and update as illustrated in Fig. 1. By reconfiguring the RF front-end according to the corresponding range limit which set by the ADC parameters and chirp characteristics, the proposed system achieves energy-efficient and scalable range adaptation.

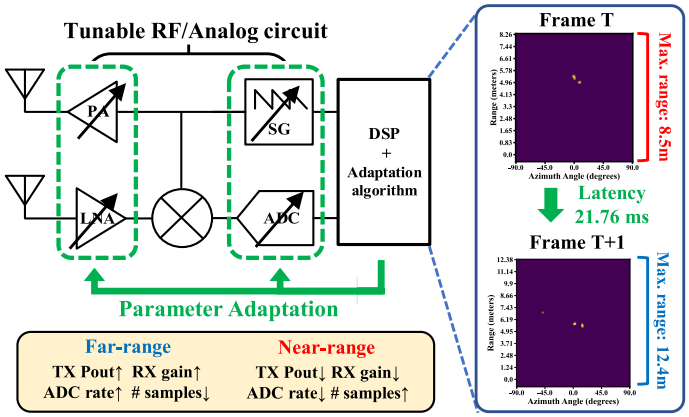


Fig. 1. Overview of the proposed range-adaptive FMCW Radar architecture. The system employs a tunable RF/analog front-end that dynamically adjusts the transmitter (TX) output power, receiver (RX) gain, ADC sampling rate and number of samples according to target range.

Utilizing the reconfigurability of the commercial radar module, we implemented the proposed framework with a median latency 21.76 ms, achieving a 33 to 69% reduction in energy consumption while maintaining robust detection performance and meeting the latency constraints.

II. Hardware Characterization

We conducted real-world experiments leveraging the tunability of Texas Instruments' AWR1843BOOST [12], as shown in Fig. 2. By adjusting the RF front-end and ADC parameters, we analyzed target detection performance across various distances and configurations.

As shown in Fig. 2(b) and Fig. 2(d), when the configured unambiguous range was insufficient, targets located beyond that limit could not be detected which reflects the limitation of the baseband processing stage. Furthermore, as illustrated in Fig. 2(c), when the RF front-end performance degraded, even targets within the unambiguous range failed to be detected.

These observations lead to the conclusion that by aligning the computation-domain unambiguous range with the RF front-end-domain detectable range, we can identify an optimal operating "sweet-spot" that minimizes energy consumption while maintaining reliable detection. Moreover, by using a single radar module with dynamically adjustable parameters, the system can modify its unambiguous range limit according

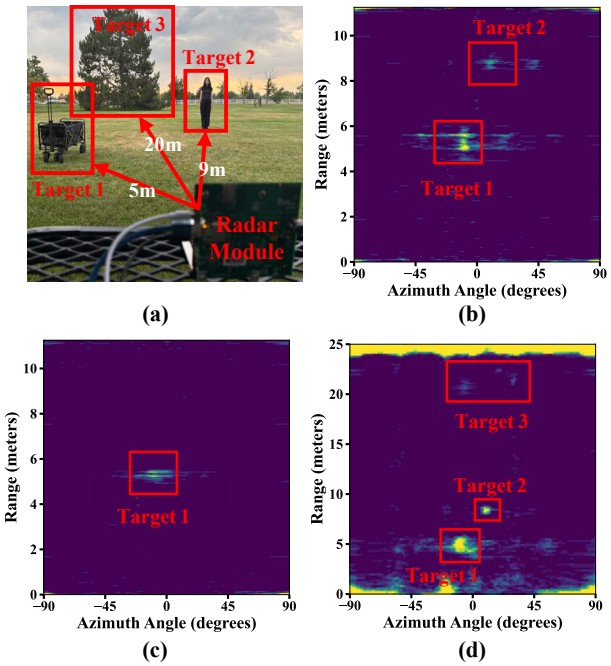


Fig. 2. Impact of RF front-end and ADC adaptation on range-angle mapping performance. (a) Experimental setup showing three targets at 5m, 9m, and 20m from the radar module. (b) Baseline setup with high TX power, RX gain, and 11.25m unambiguous range. (c) Degrade RF front-end performance limits detection to the 5m target. (d) Increasing the ADC sampling rate extends the unambiguous range to 25m, enabling detection of all three targets.

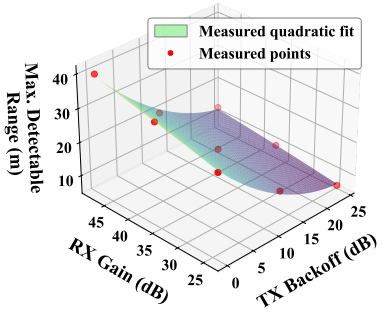


Fig. 3. Characterized maximum detectable range versus TX backoff and RX gain. The measured points were fitted with a quadratic surface to model the RF front-end behavior of the radar module for adaptive operation.

to environmental conditions and thereby expand the visible range of a single radar unit.

We experimentally characterized the RF front-end hardware performance of the AWR1843BOOST by testing nine combinations of TX transmit power and RX gain. For each configuration, the maximum detectable range was measured, used to fit a quadratic surface model as shown in Fig. 3, which served as the basis for subsequent RF front-end adaptation. The ADC sampling rate, number of ADC samples, and chirp slope were also verified within their hardware-controllable limits through experiments. Table 1 summarizes the fixed parameters used in this work, while Table 2 lists the adaptive parameters, which are dynamically configured based on the detected target range.

Table 1. Fixed Radar Parameters Used in This Work

Parameter	Value
Carrier frequency	77 GHz
TX / RX	3 / 4
Chirps	32
Frame duration	100 ms

Table 2. Parameter Selected for Adaptation and the Ranges of Values

Parameter	Value
TX Pout backoff (P_{TX})	0 to 12 dBm
RX gain (G_{RX})	24 to 48 dB
ADC sampling rate (f_s)	3 to 5 MSPS
ADC samples (N_s)	64 / 128 / 256
Chirp slope (S)	30 to 100 MHz/μs

III. Adaptation Framework

The proposed framework begins with the raw ADC cube captured by the radar module, as illustrated in Fig. 4. In the signal processing pipeline, the system performs Range FFT, Doppler FFT, static clutter removal, Capon angle-of-arrival (AoA) estimation [14], and 2D Ordered-Statistics (OS) CFAR [15] algorithm to extract range-angle information. Capon AoA and OS-CFAR are adopted to enhance angular and range selectivity without requiring a large number of FFT bins, ensuring reliable detection accuracy even as the ADC rate and corresponding FFT bin resolution vary. The core of the framework is the adaptation pipeline, which executes as a sequential, five-stage process:

Detection Filtering and Weighting: The pipeline ingests the current detections ($\mathcal{D}_t = \{(R_i, w_i)\}_{i=1}^{N_t}$) and applies an outlier filter to discard spurious reflections near the previous range limit ($r_{\max}^{t-1}$).

Estimate Spatial Uncertainty: This stage computes a stable representative distance, $R^\star$, using a weighted-percentile operator. This involves sorting detections by R_i, normalizing weights so ($\sum w_i = 1$), and finding the distance R_k corresponding to the smallest index k whose cumulative weight meets a high-percentile threshold ($q \in [0.8, 0.95]$), satisfying $\sum_{i=1}^{k} w_i \geq q$.

Refine Next Frame Range: A headroom margin is added to $R^\star$ and the result is clamped within the hardware's allowable bounds. This step yields a new target range R_{tgt} and avoids spurious detections frequently observed near the zero and maximum range edges of the range-angle map.

Temporal Stabilization: To avoid abrupt configuration changes, this stage applies an exponential moving average (EMA) with coefficient γ and a change limit Δ_{max} to R_{tgt}, producing the final target range r_{max}^t for the next frame.

Parameter Selection: Finally, the system queries a pre-trained ML model [13] with the r_{max}^t input. The model outputs the five hardware parameters, $\theta^t = (P_{TX}, G_{RX}, N_s, f_s, S)$, that satisfy $\mathrm{MUR}(\theta) \geq r_{\max}^t$ while minimizing total energy (modeled in Sec. IV), generalizing better than a simple LUT.

The chosen configuration θ^t is deployed only if it differs from the current one. Since f_s and N_s vary, the FFT size and range-bin spacing are automatically adjusted. This realizes a

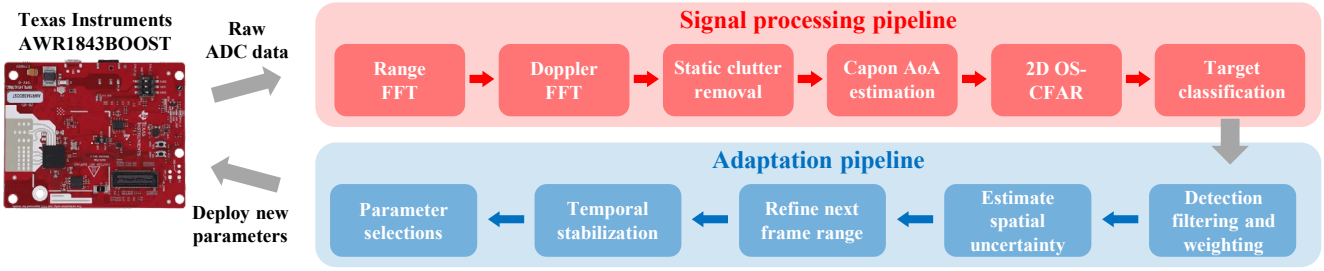


Fig. 4. Closed-loop signal processing and adaptation pipeline of the proposed FMCW radar framework. The signal processing chain extracts range-Doppler-angle information of the target. The adaptation pipeline refine next frame range limit and updated parameters which mapped via a pre-trained ML model [13] under an energy-cost constraint and deployed back to the radar hardware.

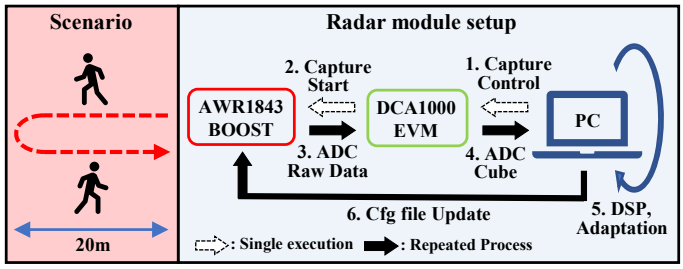


Fig. 5. Measurement scenario and real-time experimental setup

cross-domain adaptation (RF, analog, digital) that maintains detection quality while achieving significant power savings.

IV. Experimental Results

To analyze the energy reduction achievable through both the RF/analog circuitry and baseband processing, we model the total energy consumption E_{total} as:

$$E_{total} = E_{RFFE} + E_{ADC} + E_{PROC} \tag{1}$$

by decomposing it into three components. Each term is parameterized by controllable hardware and algorithmic variables to capture the impact of parameter adaptation on power efficiency. The E_{RFFE} term is an empirical formula derived from hardware characterization, which uses the logarithmic control values as direct numerical inputs: $E_{RFFE} = (C_{base} - P_{TX} + \beta \cdot G_{RX}) \cdot \frac{N_s}{f_s}$ where P_{TX} is the TX backoff value, G_{RX} is the RX gain value, and C_{base} (e.g., 12 in our model) and β are fitted coefficients. The remaining components are modeled as: $E_{ADC} = \alpha \cdot C \cdot V^2 \cdot N_s$ and $E_{PROC} = \text{FLOPs} \cdot (10\ \text{nJ/FLOP})$ where E_{ADC} is parameterized by α, capacitance C, voltage V, and number of samples N_s. The FLOPs for E_{PROC} include both signal and adaptation processing.

Fig. 5 presents the measurement scenario and real-time experimental setup. The PC and DCA1000 are connected via an Ethernet interface, while the AWR1843BOOST and DCA1000 are linked through an LVDS cable. Baseband processing are executed on a 12th Gen Intel(R) Core(TM) i5 CPU. Further implementation details of the data-capture can be found in [13], [16]. Based on this setup, we conducted a simple outdoor experiment with a stationary radar module. A human

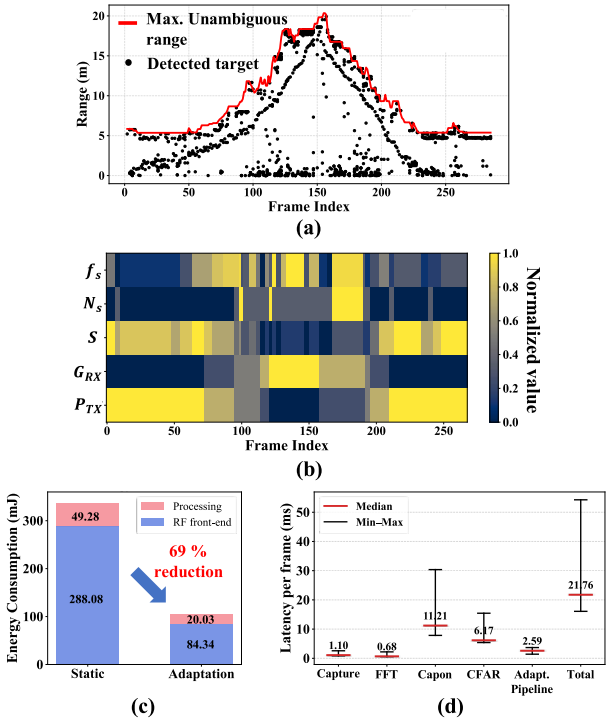


Fig. 6. Measurement results. (a) Detected range while the person moves from 0 m to 20 m back with dynamic unambiguous range adaptation. (b) Parameter variation across 250 frames. (c) Energy consumption across 250 frames. (d) Average per-frame latency of each processing stage.

walked away from the radar to 20 m and then returned to the starting point, observe how the MUR and target detections are changed under the proposed adaptive framework.

The results of real-time experiment are summarized in Fig. 6. Fig. 6(a) shows the detected range trajectory of a walking human target, where the black dots represent detection results after OS-CFAR and the red line indicates the dynamically adjusted MUR. This demonstrates that the proposed adaptation framework can change the range configuration in real-time while continuously tracking the target, thereby extending the detectable range coverage using a single radar module.

Fig. 6(b) visualizes the normalized variation of parameters over frame, showing how the system adjusts each parameter in response to the target's distance.

Fig. 6(c) presents the energy reduction achieved by

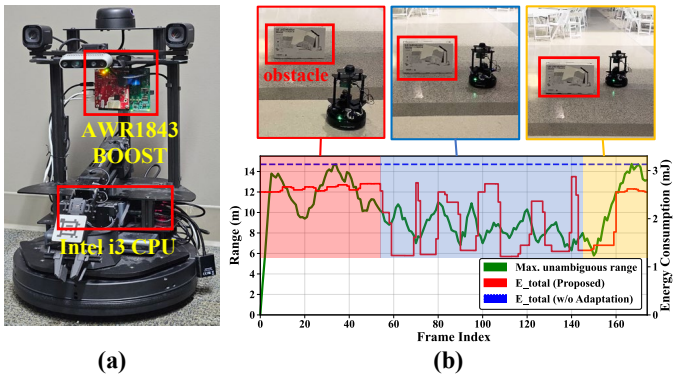


Fig. 7. Demonstration of the proposed adaptive framework on a mobile robot. (a) Hardware setup. (b) Navigation scenario with adaptive maximum unambiguous range and energy consumption.

adaptively optimizing the RF front-end performance with respect to range limit. Energy values computed using the proposed model (1) indicate a 69% reduction under adaptive operation, confirming that dynamically matching the sensing range to application requirements yields substantial energy savings.

Fig. 6(d) shows the per-frame latency of each processing stage. The median latency of the entire processing and adaptation pipeline is 21.76 ms. Separately, uploading the updated configuration to the radar module takes an additional 20 ms. The complete end-to-end latency thus remains well within the 100 ms frame duration used in this work, demonstrating that stable real-time adaptation is achievable.

V. Extra Demonstration

We validated the system on a LoCoBot [17] equipped with an AWR1843BOOST radar and an Intel i3 edge processor (Fig. 7(a)). Using only our adaptive radar sensing pipeline, the robot successfully navigated its environment. As shown in Fig. 7(b), the system shortens its sensing range when obstacles are detected and restores full range afterward, achieving a 33% reduction in energy consumption relative to a non-adaptive baseline while preserving reliable obstacle detection.

VI. Conclusion

This paper introduces a real-time radar adaptation framework that reconfigures sensing parameters to focus on relevant targets and reduce energy use. A lightweight spatio-temporal algorithm enables fast, stable updates with preserved context. Real-world demonstrations with a commercial FMCW radar show 33–69% energy savings with robust detection and single-frame timing. The results establish a practical baseline for future cognitive radar and highlight opportunities for intelligent, energy-aware operation.

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